CSci 135: M1 →

Software Engineering Concepts

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Version: v5.2026

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Course Outcome

Explain the concepts, principles, and professional context of software engineering.

Learning Outcome

Discuss the importance and relevance of software engineering.

Outline

1. Software Development
2. Lifecycle Models
3. Methodologies
4. Software Engineering
5. Software Engineering Layers
6. Quality in Mind

Software Engineering Concepts

Software Development

What is Software Development?

The ***entire process*** from the initial ideation to the maintenance of a software.

* It requires skills on programming languages, libraries, frameworks, and relevant tools.



Software Development Lifecycle

It begins with the decision of ***starting a software project until delivery*** (IEEE-CS, 2014).

* The distinct phases have no specific order but often overlap or run in parallel.

Planning

Implementation



Analysis

Testing

Maintenance



Design

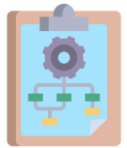
Deployment

Software Engineering Concepts

Lifecycle Models

Lifecycle Model

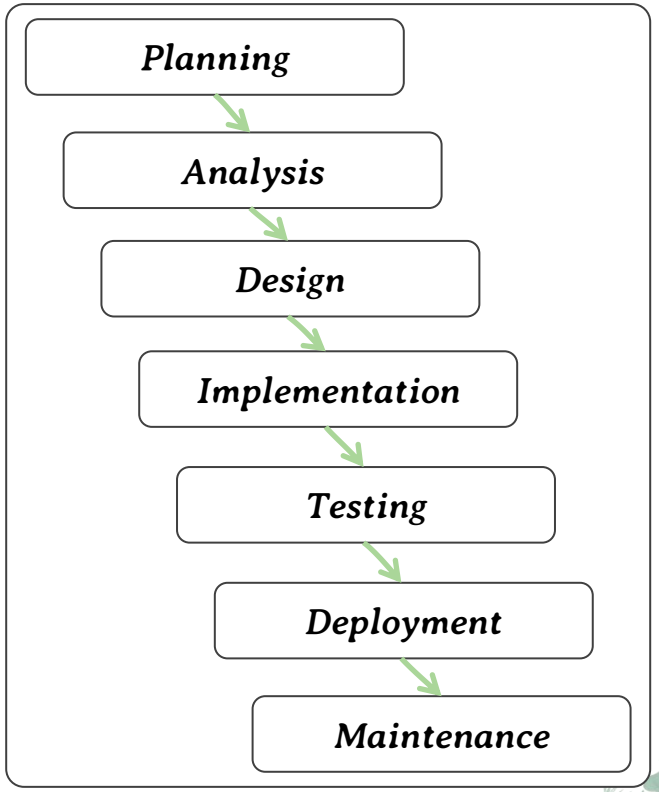
It provides the ***ordering of the phases*** of software development.



Waterfall Model

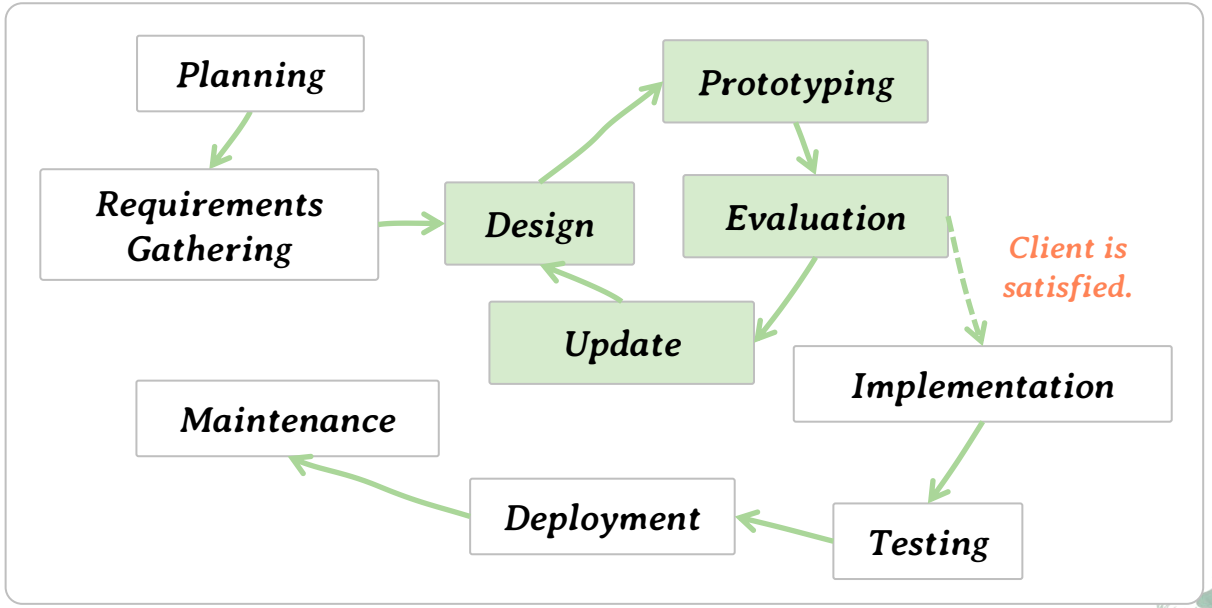
Software development done sequentially.

* Ideal to small software development done by those outside the field of computing; ***implementation*** here means the conversion of design to actual code, while development is the whole planning to maintenance phase.



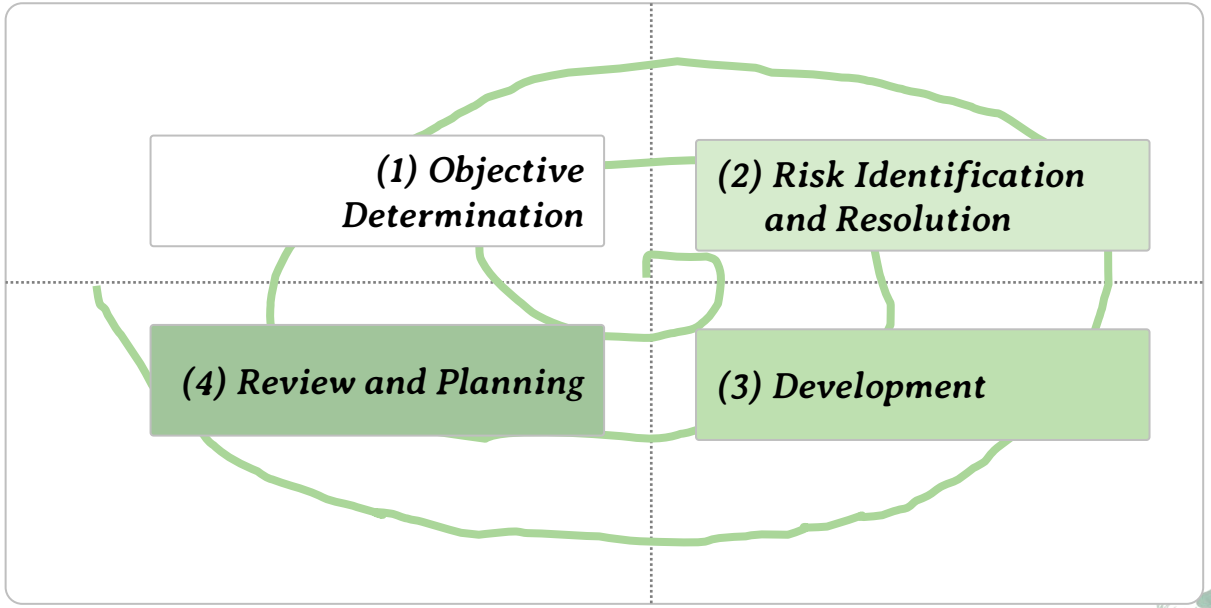
Prototype Model

It creates a preliminary version of the software before the final development.



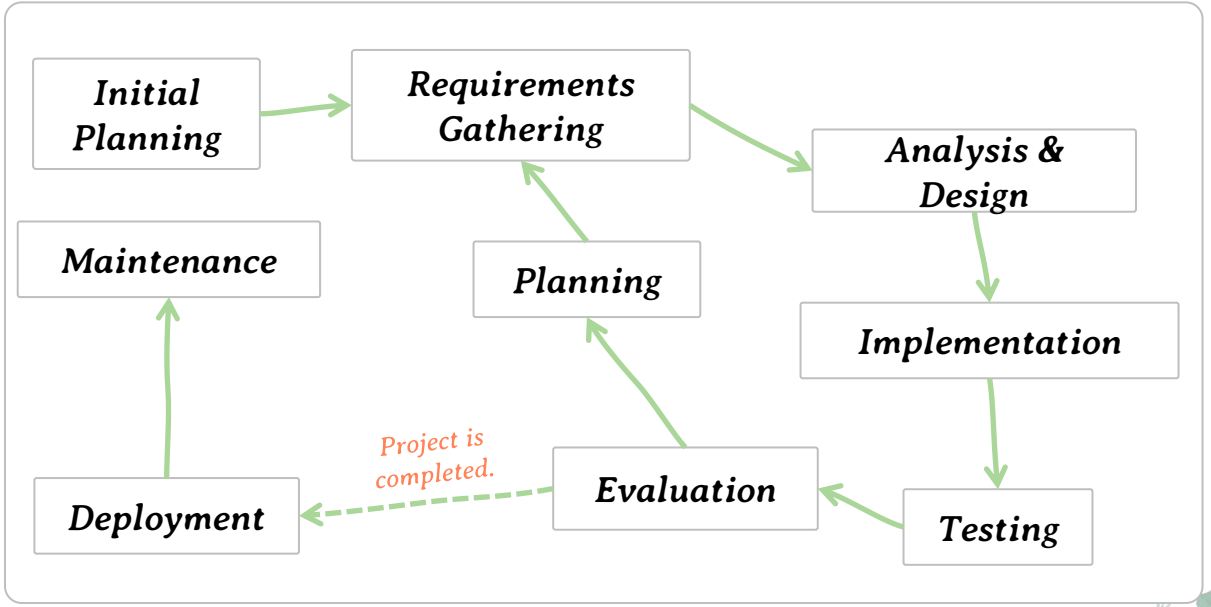
Spiral Model

It is a combination of ***sequential and prototype*** models, followed in a more defined manner.



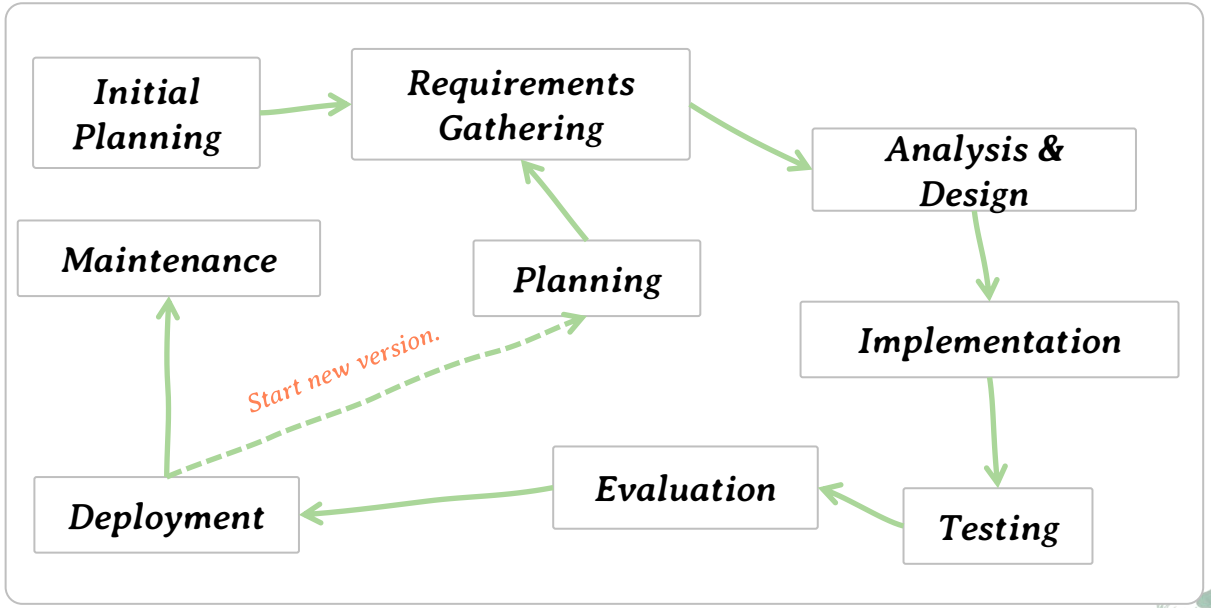
Iterative-Incremental Model

It starts with a not fully usable ***basic implementation*** then adds remaining features in each iteration until completed.



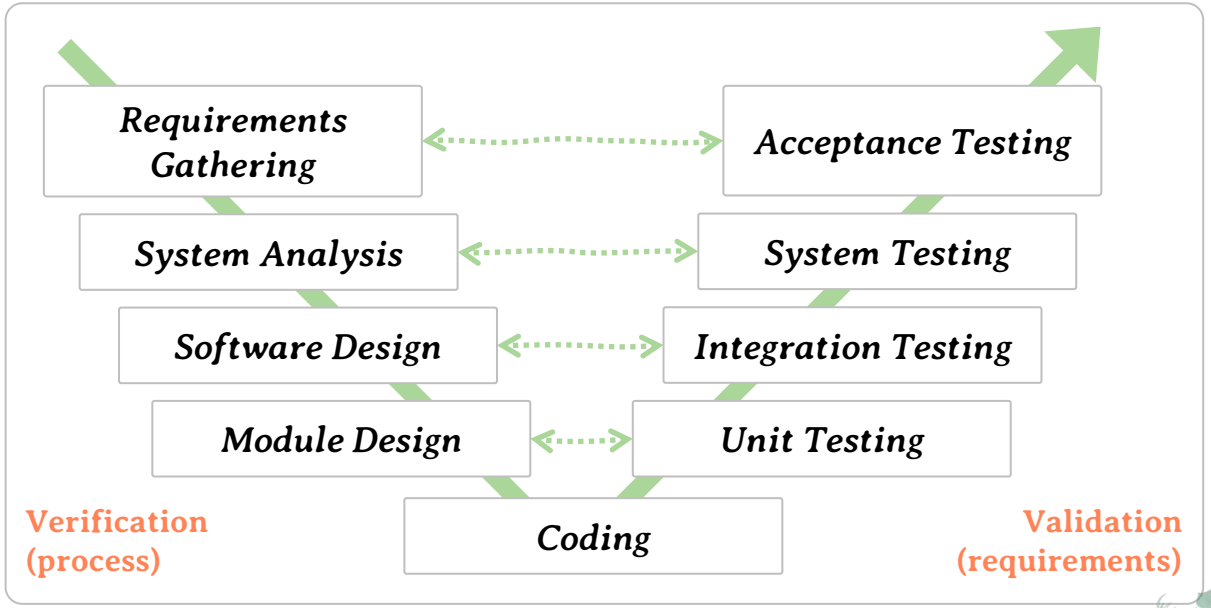
Evolutionary Model

It is like iterative-incremental model but ***each iteration is a complete version.***



V Model

It emphasizes on ***verification and validation*** through a structured approach; the output in the left is the input at the right.



Software Engineering Concepts

Methodologies

SW Development Methodology

It is a formal and granular approach with the aim of reusability following a ***standardized set of terminologies, phases, deliverables,*** and ***procedures*** for software development.

SW Development Methodology

#1 Structural Methodology

It focuses on business activities following ***traditional principles***.

Principles	Characteristics	Major Deliverables
Structuralism	*Strategy: Divide and Conquer*	*Analysis Report*
Stepwise Refinement		*Data Flow Diagram*
Modularization	*Modules*	*Structural drawings*
Abstraction	*Logic*	*Specification*

Read More: Structured Analysis and Structured Design SA/SD

SW Development Methodology

#2 Information Engineering

It is ***data-focused*** and with strong understanding on ***data***, ***business systems***, and ***information technology***.

Principles	Characteristics	Major Deliverables
Strategy Plan	*Organizational Perspective*	*Analysis Report*
Area Analysis	*Enterprise-level Data Models*	*ERD*
System Design		*Database Schema*
System Development	*Data-based Logic*	*Specification*

SW Development Methodology

#3 Object-Oriented Methodology

It focuses on the identification of ***objects***, ***classes***, and ***its relationships*** through a design model.

Principles	Characteristics	Major Deliverables
Requirements	*Data-Logic Integration*	*Conceptual Diagram*
Analysis and Design	*Objects*	*UML Diagrams*
Testing	*Advanced Modularization and Inheritance*	
Deployment		

SW Development Methodology

#4 Agile Methodology

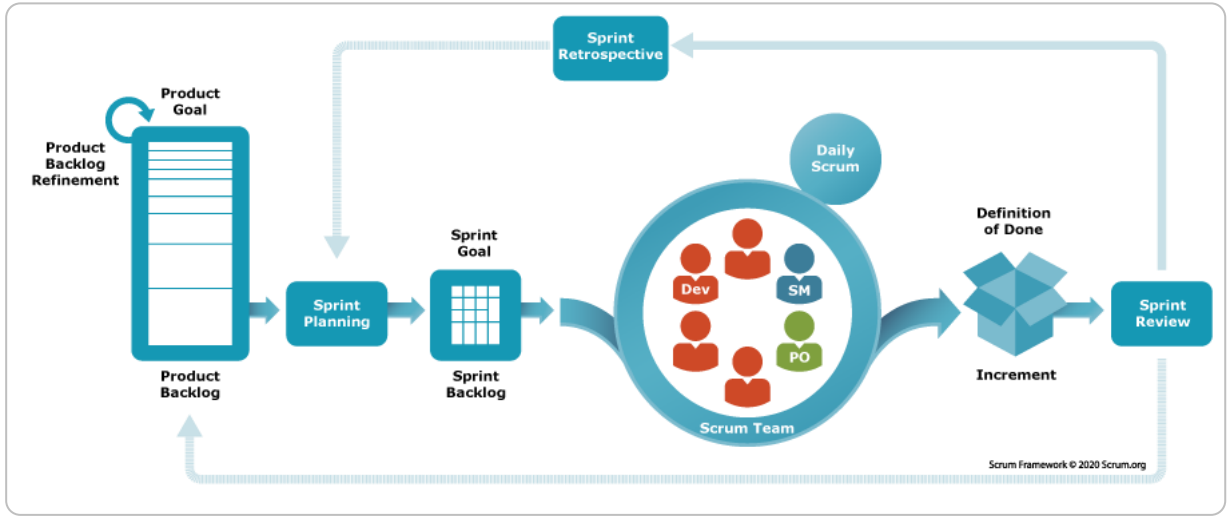
It's a framework that customizes its methods for each project, following an ***iterative-incremental development*** approach where each iteration produces a ***shippable product***.



SW Development Methodology

#5 Scrum Framework

It is a form of Agile methodology designed for ***project management in mind.***



Read More: https://www.scrum.org/resources/what-scrum-module

Software Engineering Concepts

Software Engineering

What is Software Engineering?

It is the application of a ***systematic, disciplined, quantifiable approach*** to the development, operation, and maintenance of software; that is, the ***application of engineering*** to software development (IEEE, 1990).

What is Software Engineering?

It is the establishment and ***use of sound engineering principles*** to economically obtain software that is reliable and works efficiently on real machines (Bauer, 1972).



Sound Engineering Principles

- Going beyond software development...
  - Cover areas such as in ***requirements development***, ***UI/UXD***, ***systems engineering***, and ***Project Management***, among others.
  - **Goal:** ***overall efficiency*** and ***higher success rate***.

Economical Software

- Business as usual...
  - Optimized processes increase profitability.
  - **Goal**: *Cost-effective* deliverables.

Works on Real Machines

- Working product...
  - Economics of computational and memory resources.
  - Efficient and performant software:
    - Rigorous security, performance, stress, and reliability testing.
  - **Goal**: ***Customer satisfaction.***

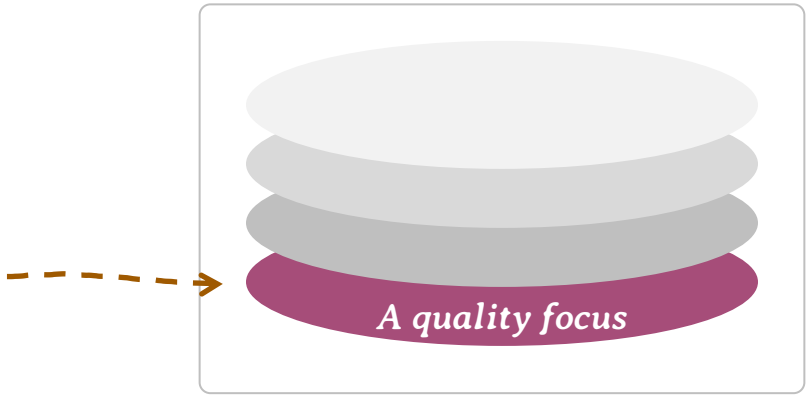
Software Engineering Concepts

Software Engineering Layers

Software Engineering Layers

#1 **A quality focus**

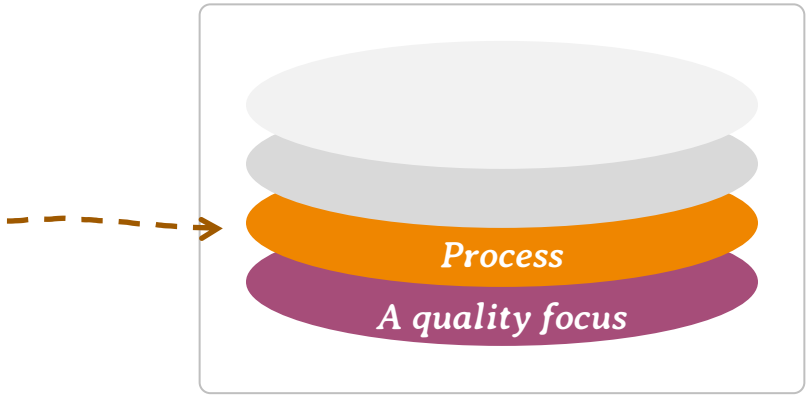
It refers to the ***continuous process improvement*** principles detailing data integrity, data security, and software usability and maintainability.



Software Engineering Layers

#2 **Process**

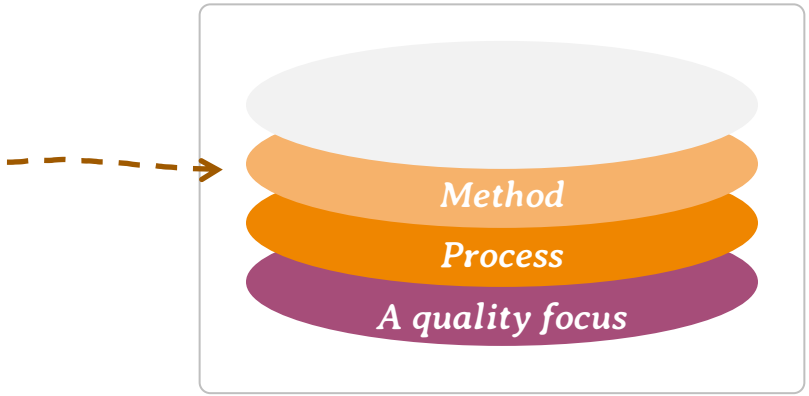
It forms the basis for ***management of control*** of all activities, actions, and tasks for the efficient delivery of software.



Software Engineering Layers

#3 **Methods**

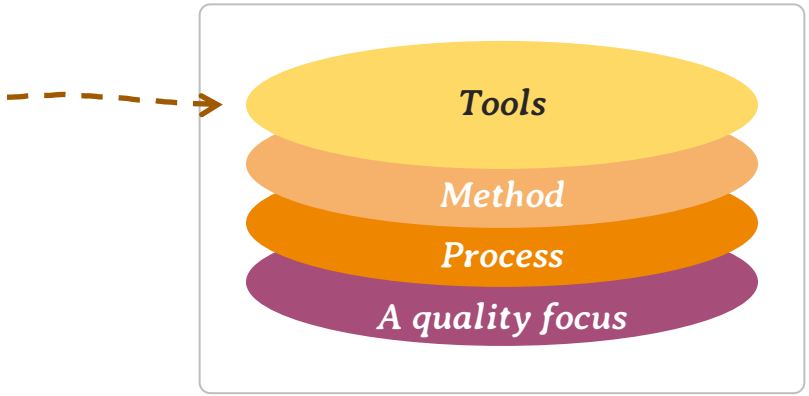
It encompasses a broad array of tasks including communication, requirements analysis, design modelling, program construction, testing and support.



Software Engineering Layers

#4 **Tools**

It is a ***set of tools*** for automated and semi-automated processes, so that the output of the previous is a valid input to another.

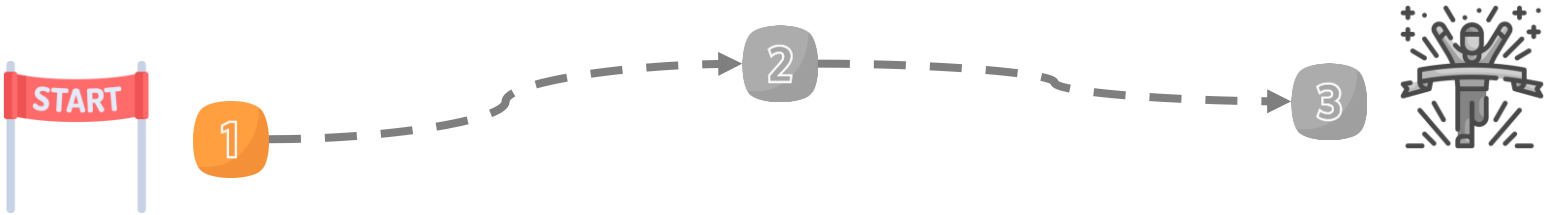


Software Engineering Concepts

Quality in Mind

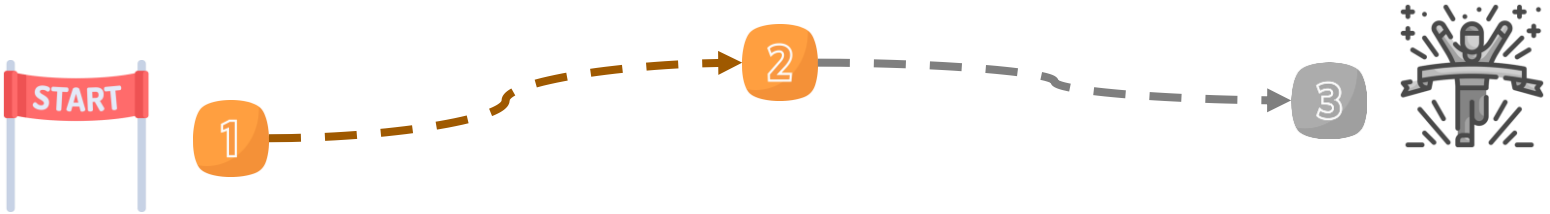
“Quality in Mind” Compass

- **STEP #1:** Are we being ***efficient and effective*** in all aspects of the project? (***Yes***/No)



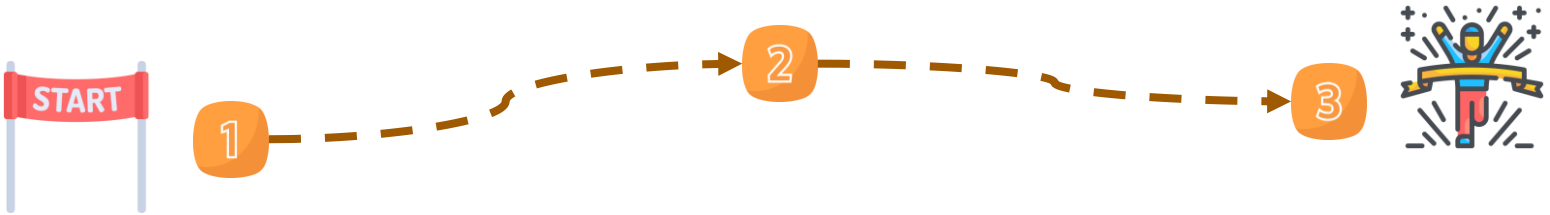
“Quality in Mind” Compass

- **STEP #2** Are we managing our ***limited resources*** well without compromising quality? (***Yes***/No)



“Quality in Mind” Compass

- **STEP #3** Will our software **run smoothly** when used in production? (***Yes***/No)



Summary

- ***Software Development*** is entire process of ideation to the maintenance of a software.
- A ***Lifecycle Model*** provides the ordering of the phases of software development.
- A ***Software Development Methodology*** is a standardized approach to building software.

Summary

- ***Software Engineering*** is the application of relevant approaches to efficiently deliver high-quality software.
- The ***Software Engineering Layers*** illustrate software built through continuous process improvement, task management, and tool usage.
- The ***"Quality in Mind" compass*** is a mental framework for managing tasks and resources to ensure the delivery of high-quality software.

Scan me!

Evaluation Form

forms.gle/eUbH7NiBsvQGaPt27

Thank You!

References

- IEEE. (1990). IEEE Standard Glossary of Software Engineering Terminology. IEEE Std 610.12-1990, 1–84. IEEE Std 610.12-1990. https://doi.org/10.1109/IEEESTD.1990.101064
- IEEE Computer Society. (2014). Guide to the Software Engineering Body of Knowledge (SWEBOK®): Version 3.0 (P. Bourque & R. E. Fairley, Eds.). IEEE Computer Society Press.